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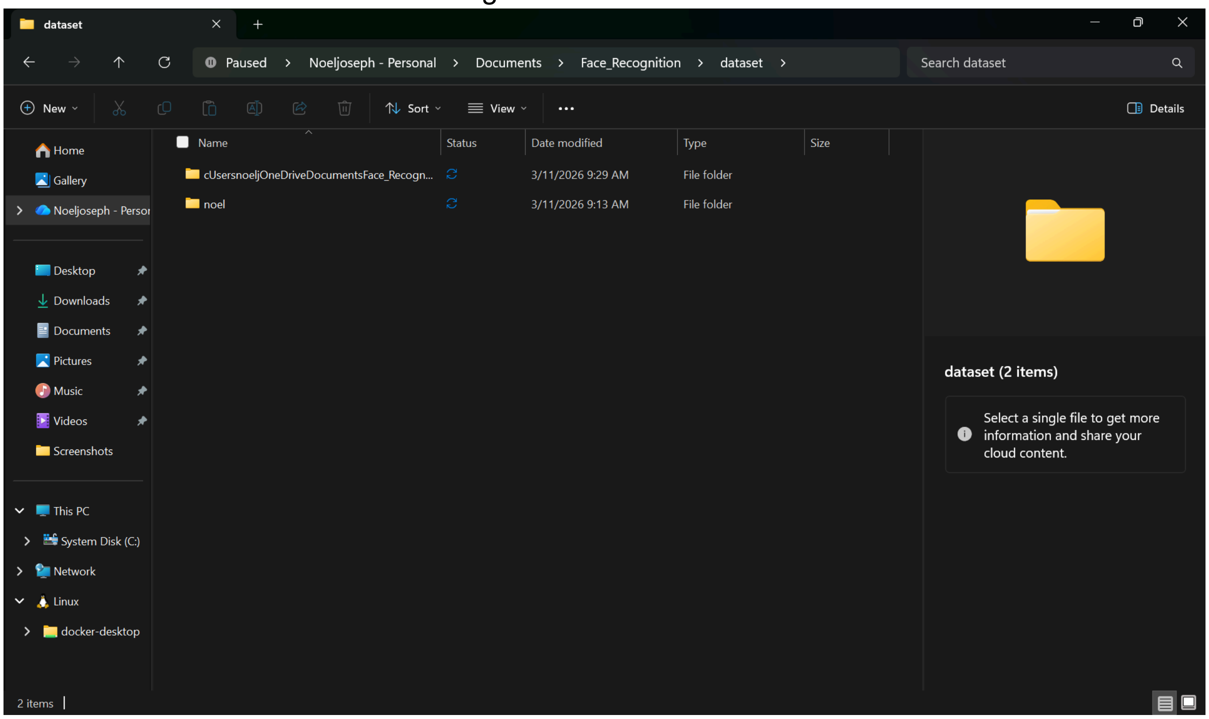
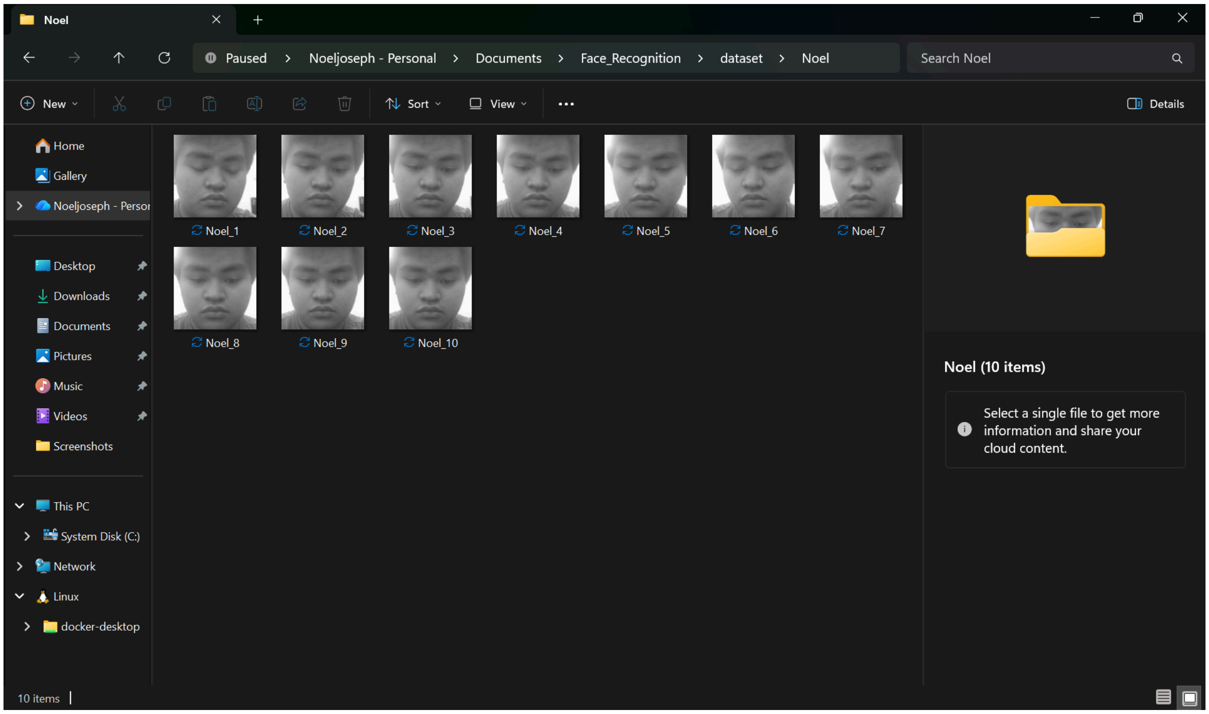
IT22S3	3/11/2026

1. Screenshot of dataset folder with images





2. Screenshot of training terminal output

```
(.venv) PS C:\Users\noelj\OneDrive\Documents\Face_Recognition> & C:/Users/noelj/OneDrive/Documents/Face_Recognition/.ve
nv/Scripts/python.exe c:/Users/noelj/OneDrive/Documents/Face_Recognition/FaceRec_Lab/collect.py
(.venv) PS C:\Users\noelj\OneDrive\Documents\Face_Recognition> & C:/Users/noelj/OneDrive/Documents/Face_Recognition/.ve
nv/Scripts/python.exe c:/Users/noelj/OneDrive/Documents/Face_Recognition/FaceRec_Lab/collect.py
Enter your name: Noel
Enter number of images to capture (recommended 30): 10

Press SPACE to save an image. Press Q to quit.

Saved dataset\Noel\Noel_1.jpg
```

3. Screenshot of recognition working

```
(.venv) PS C:\Users\noelj\OneDrive\Documents\Face_Recognition> & C:/Users/noelj/OneDrive/Documents/Fa
nv/Scripts/python.exe c:/Users/noelj/OneDrive/Documents/Face_Recognition/FaceRec_Lab/collect.py
Saved dataset\Noel\Noel_6.jpg
Saved dataset\Noel\Noel_7.jpg
Saved dataset\Noel\Noel_8.jpg
Saved dataset\Noel\Noel_9.jpg
Saved dataset\Noel\Noel_10.jpg
Data collection complete. 10 images saved for Noel.
(.venv) PS C:\Users\noelj\OneDrive\Documents\Face_Recognition>
```

4. Short screen recording (1–2 minutes)
5. Reflection paper

“This Activity consists focuses on getting facial recognition using python and storing the local images in your local files”

6. GDRIVE LINK - Lab 10 Activity-HCI 2

- Why collect multiple images per person?

“Facial recognition relies on variation in facial features such as different angles, expressions and micro-changes.”

- What happens if lighting conditions change?
  “Lighting affects how the camera perceives facial features.”
- What does the confidence value represent?
  “It’s a numerical measure of similarity between the input face and stored templates.”
- Why is LBPH considered lightweight compared to deep learning models
  “LPBH or Local Binary Patterns Histograms uses simple texture descriptors rather than millions of parameter that requires massive datasets or GPUs”
- How can this system be improved for real-world deployment?
  “Data Augmentation simulates variations in lighting, pose, and occlusion. And continuous updates on the dataset as students grow and change their appearance and UI. ”
- Ethical concerns of using facial recognition in schools
  “This concern stems from violating one’s privacy. Storing biometric data of minor raises serious concerns ”